INT. FILMCROWD LABS - DATA SCIENCE - DAY

Greenlight Score

Project brief: predicting a film's investment case from its script

PREPARED FOR	FilmCrowd data science team
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SECTION 01

Project overview

Build a model that reads a film script and tells an investor how likely the film is to make money, how much, and why. The output is an investor view, not a single quality score.

The question we answer. Why should an investor put money into this film? An investor asks four things: will I get my money back, how much could I make, how much could I lose, and why should I believe the answer.

Approach. LLMs read the script and estimate everything a script can reveal. Internet data adds current market trends. FilmCrowd's artist interviews become a proprietary scoring rubric. Our 1M movie dataset supplies comparable films and the outcomes the models learn from. A small neural network combines everything. LLMs read and explain; trained models predict.

In scope

- One script in, one investor view out
- Five expert sub-scores (script, market, interview, comps, and talent in version 2) plus a combiner model
- A budget estimate derived from the script
- Emotion, energy and genre-mix curves read by a 1D CNN
- Simulated audience screening and a written report with scene-level fixes

Not in scope: marketing spend, which a script cannot show, and analysis of rough cuts or video.

SECTION 02

Inputs

At prediction time the only required input is the script. Everything else is reference data we hold or optional context the user adds.

Input	What it is	When it is used	Required
Script	Screenplay as PDF, Final Draft or Fountain	Every prediction	Yes
Internet data	Weekly box office, release calendars, streaming top 10, trade news	Market trends at prediction time	Yes, refreshed regularly
Artist interviews	Transcripts FilmCrowd has collected	Once, to build the rubric	Yes, built once
1M movie dataset	Metadata, budget, gross, ratings, synopses	Comps and training labels	Yes
Cast options	Actors under consideration for lead roles	Talent scenarios (version 2)	Optional
Release window	Planned release date	Timing and competition features	Optional

Why the script is the only required input. FilmCrowd judges films before they exist. At that point the script is the one certain thing. The model must work without release plans or cast, and get sharper when they are given.

SECTION 03

Outputs

The model returns an investor view: a headline probability, a return range, and the reasons behind them.

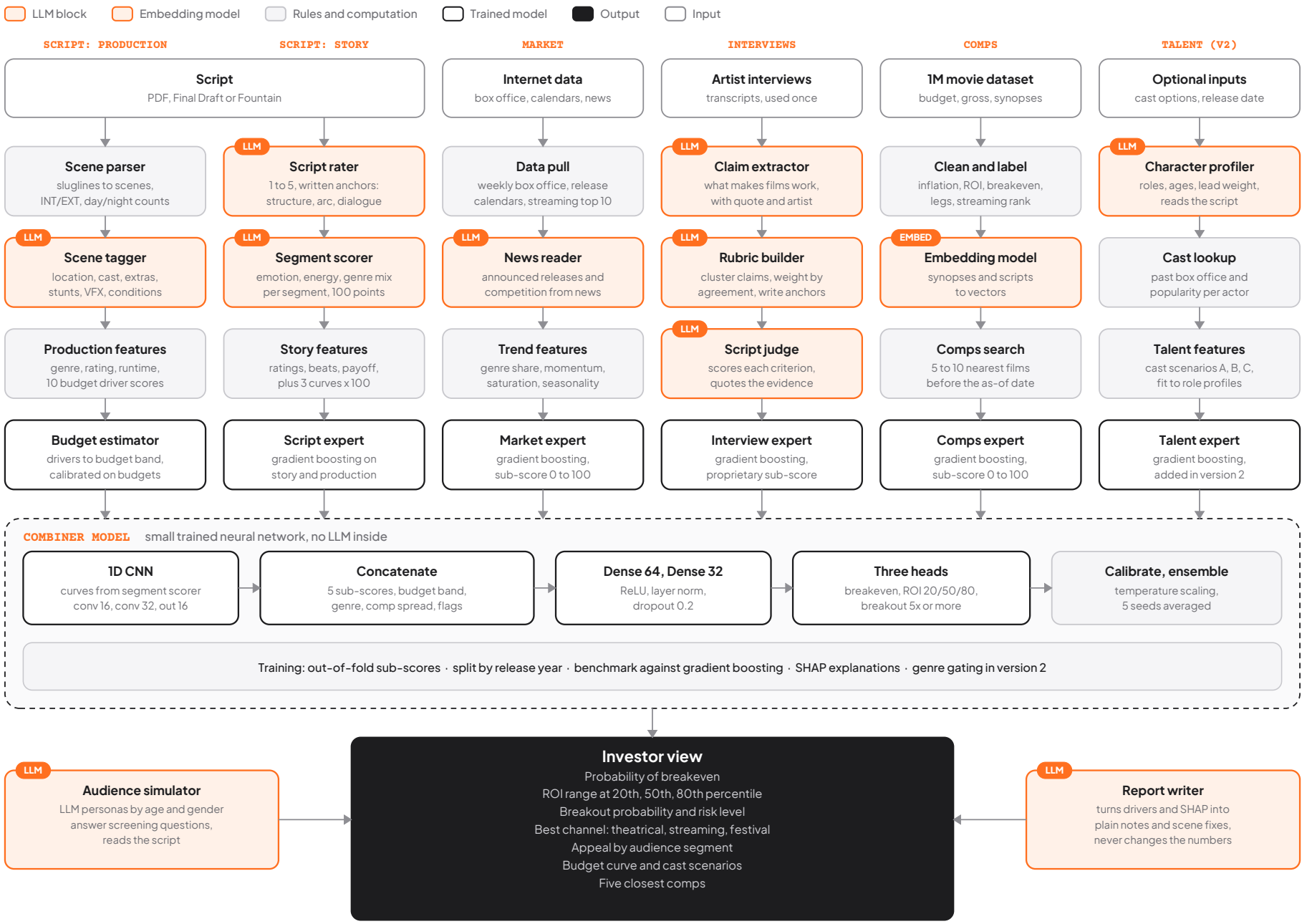
Output	What it says	Why an investor cares
Probability of breakeven	Chance the film recoups its cost (about 2.5x budget in gross)	Do I get my money back
ROI range	Return at the 20th, 50th and 80th percentile	Downside, expected case and upside in one view
Breakout probability	Chance of 5x return or more	The upside that justifies the risk
Risk level	Width of the range plus model uncertainty	Same expected return can carry very different risk
Best channel	Theatrical, streaming or festival route, with its own forecast	Many films never get a meaningful theatrical run
Segment appeal	Appeal by age and gender segment	Shows who the film is for and how big that audience is
Sub-scores	Script, market, interview, comps, talent; each 0 to 100	Where the strength or weakness comes from
Top drivers and fixes	Three factors up, three down, with scene-level notes	Explainable and actionable
Closest comps	Five similar past films and their outcomes	Investors already think in comparables
Budget curve and cast scenarios	Breakeven probability per budget band and per cast option	The same script can be a good bet at $5M and a bad one at $50M

SECTION 04

Detailed model

The next page shows the full model. Purple blocks are where an LLM is used; everything that produces a number the investor sees is a trained model (teal) or a calculation (gray). This separation is deliberate: LLMs read, tag and explain; they never set the final probabilities.

Detailed model: one script in, an investor view out



Where LLMs are used, and why

Every purple block in the diagram, with what it returns. Each LLM call uses a fixed prompt, a written rubric where it rates, and a structured JSON output so its results can be stored and checked.

LLM block	Stage	What it returns	Why an LLM
Script rater	A	1 to 5 ratings for structure, arc, dialogue, tone, payoff, with reasons	Judging craft needs reading comprehension
Scene tagger	A	Per scene: location, cast, extras, stunts, VFX, special conditions	Free text descriptions, no fixed format
Segment scorer	A	Emotion, energy and genre mix for each of 100 segments	Tone is implicit in action and dialogue
News reader	B	Announced releases and dates from trade articles	Unstructured news into structured rows
Claim extractor	C	Every success claim with quote and artist	Interviews are free conversation
Rubric builder	C	Clustered criteria, weights, 1 to 5 anchors	Grouping similar ideas phrased differently
Script judge	C	Score per criterion plus the passage that justifies it	Applying a written rubric to a script
Embedding model	D	Vectors for synopses and scripts	Similarity of stories, not just metadata
Character profiler	E	Lead roles with age, gender, screen weight	Needed to match cast options to roles
Audience simulator	F	Persona answers to screening questions, by segment	Cheap first read before real test screenings
Report writer	Output	Plain-language drivers and scene-level fixes	Turns SHAP values into notes people act on

Controls on every LLM block. Run each scoring call three times and average; flag scripts where runs disagree by more than one point. Keep a 50-script human-rated set to check LLM ratings against people. Version every prompt, since a prompt change changes the features.

SECTION 05

Stage A: script analysis

LLMs read the script scene by scene and turn it into features, a budget estimate, and three curves.

Input: the script.

Process

01 A rule-based parser splits the script into scenes using sluglines (INT. KITCHEN - NIGHT) and counts INT/EXT and day/night. No LLM needed.

02 The Scene tagger LLM tags every scene: location, speaking characters, extras, stunts, VFX, special conditions.

03 The Script rater LLM rates whole-script qualities from 1 to 5. Every level has a written anchor.

04 The Segment scorer LLM scores emotion, energy and genre mix for 100 equal segments.

05 Tags roll up into production features and ten budget driver scores; the budget estimator maps them to a band.

Output: production features, story features, a budget band (micro under $1M, low $1M to $10M, mid $10M to $40M, high $40M to $100M, tentpole $100M+), and three curves of 100 points.

Why: the script is the only thing that exists at greenlight time. Line producers already price films by breaking scripts down scene by scene, so the budget logic copies proven practice. Emotional arc shape has published links to box office.

Group	Features
Structure	Beat positions, act breaks, strength of first 10 pages, ending type
Characters	Protagonist presence, character network, agency, antagonist strength, dialogue balance
Dialogue and writing	Voice distinctness, vocabulary, humor density, overwriting
Concept	Logline clarity, originality, stakes, internal vs external conflict
Entertainment payoff	Dead stretches, setups paid off, escalation, ending satisfaction, moment density
Production	Genre, estimated rating, runtime from page count, INT/EXT ratio, day/night ratio, locations, scene count
Budget drivers	Shoot days, cast size, locations, exteriors and nights, sets, stunts, VFX, special conditions, costume and makeup, music
Curves	Emotion, energy, genre mix; 100 points each

SECTION 06

Stage B: market trends

Internet data tells us whether there is demand for this kind of film right now.

Input: the genre from Stage A, an as-of date, internet data, and optionally a release window.

Process

01 Fix an as-of date: the release date for historical films, the planned date or today for a new script.

02 Compute every feature from data before that date only.

03 The News reader LLM turns trade articles into a list of announced releases for the competition features.

04 With no release window given, run the script across every window and show the best one.

Output: genre share, genre momentum, genre saturation, recent genre ROI, seasonality index, competition (optional), streaming demand.

Why: the same script can succeed or fail depending on what audiences want when it lands. The as-of rule stops the model from seeing the future.

SECTION 07

Stage C: proprietary interview criteria

FilmCrowd's artist interviews become a scoring rubric no competitor has, and every script is judged against it.

Input: interview transcripts (once) and the script (every prediction).

Process

01 The Claim extractor LLM pulls every claim of the form "X makes a film work" or "X kills a film", with quote and artist.

02 The Rubric builder LLM clusters similar claims into criteria, weights each by how many artists raised it, and writes 1 to 5 anchors. A person reviews the rubric before use.

03 For every script, the Script judge LLM scores each criterion and quotes the passage that justifies it.

Output: an interview fit vector with evidence quotes.

Why: practitioner knowledge of what makes films work is not in any public dataset. Evidence quotes make every score checkable.

Proof requirement. Train the full model with and without the interview score. If accuracy improves, the proprietary claim is proven. This also shows which artist beliefs hold up in the data.

SECTION 08

Stage D: comps

The movie dataset finds the past films most like this script and reports how they performed.

Input: the script embedding, genre and budget band, and the movie dataset.

Process

01 Prepare the dataset once: inflation adjust, compute labels, embed every synopsis and available script.

02 Find the 5 to 10 nearest films by embedding similarity, filtered to genre and budget band.

03 Only films released before the as-of date count; a film is never its own comp.

Output: comp ROI average and spread, closest similarity (originality), comp recency, number of close comps.

Why: producers and financiers already judge projects by comparables, and comps make the prediction explainable.

SECTION 09

Stage E: talent scenarios (version 2)

Talent is the largest gap against commercial tools. We add it as scenarios rather than a requirement.

Input: the script and optional cast options from the user.

Process

01 The Character profiler LLM lists lead roles with age, gender and share of screen time.

02 For each cast option, look up the actors' past box office and popularity from public data.

03 Score fit between each actor and role, then re-run the full model per scenario.

Output: a talent sub-score and a breakeven probability for each cast scenario.

> **Why:** cast drives financing decisions. Showing scenarios side by side answers the investor's question "what if we cast X instead?"

SECTION 10

Stage F: simulated audiences

Input: the script.

Process

01 Define personas per segment: age band, gender, viewing habits.

02 The Audience simulator LLM gives each persona a scene-level summary and asks fixed screening questions.

03 Aggregate answers per segment.

Output: segment appeal scores and screening-style answers.

> **Why:** a fast, cheap first read on who the film is for. It is a feature and a report element, never a substitute for real screenings. Validate it against real audience scores before relying on it. The Movie Review app's screening survey cards are the natural front end.

SECTION 11

Success labels

Success is measured after release, so labels come from the movie dataset, never the script.

Label	Definition	Used for
Breakeven	Worldwide gross at or above 2.5x budget	Breakeven head
Log ROI	log(worldwide gross / budget), inflation adjusted	ROI range head

Label	Definition	Used for
Breakout	ROI of 5x or more	Breakout head
Streaming success	Top 10 chart weeks or hours viewed, as a percentile	Streaming channel forecast
Legs	Total gross / opening weekend	Satisfaction signal
Audience score	Audience rating, percentile within year and genre	Satisfaction signal, simulator validation

Why 2.5x. Theaters keep roughly half the gross and marketing costs extra, so a film typically needs about 2.5x its production budget to recoup. It is a rule of thumb and can be tuned.

SECTION 12

Expert models

Each stage feeds its own gradient boosting model. Each expert's prediction, converted to a 0 to 100 percentile against past films, is its sub-score. A script score of 82 means: on script alone, this looks better than 82% of past films.

Why learned sub-scores. A hand-made rating has no meaning by itself. A learned sub-score is a real prediction, tied to outcomes. The LLM ratings stay visible underneath as the explanation.

Why gradient boosting. The inputs are tables of features; gradient boosting is the strongest and most explainable choice for tabular data at this scale.

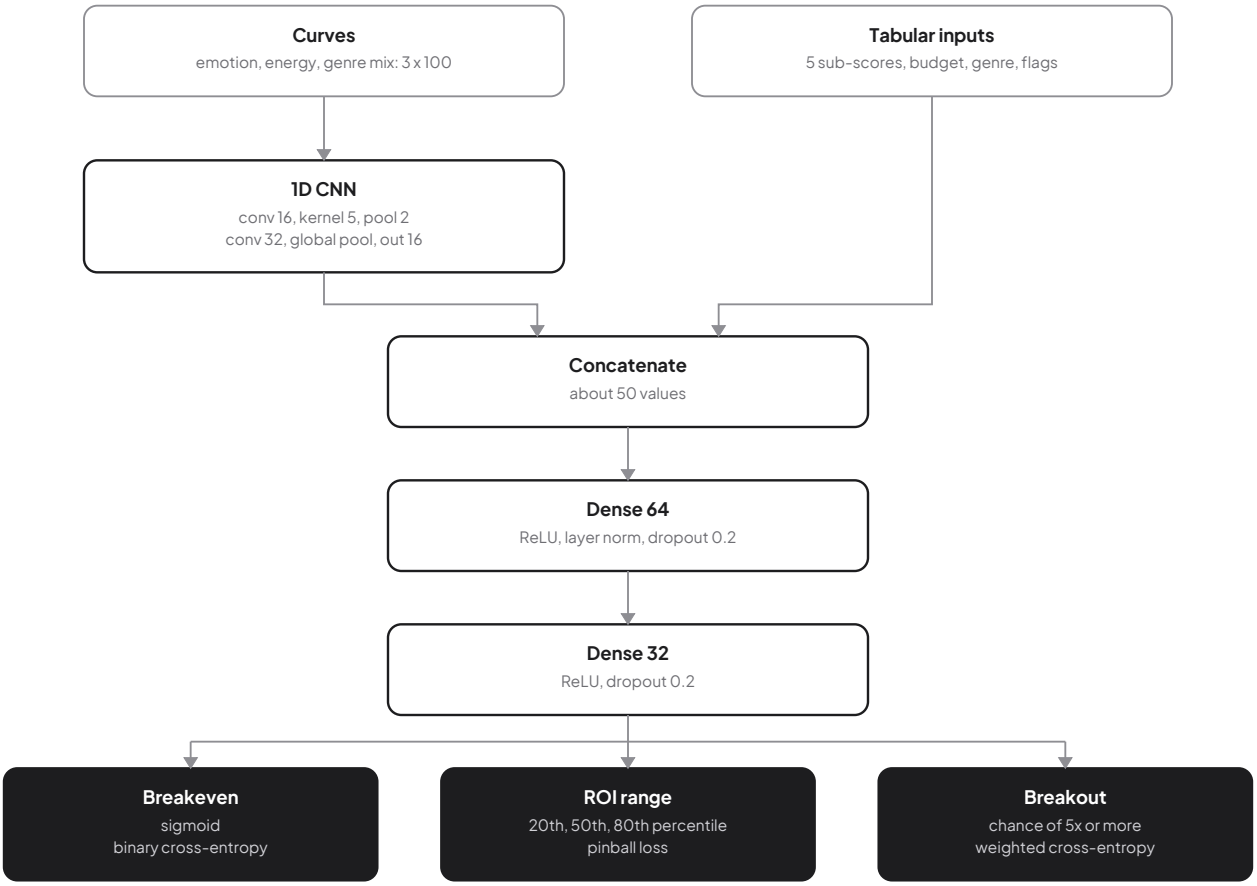
SECTION 13

Combiner model

A small multi-head network turns the sub-scores and curves into the investor view.

Combiner architecture

Two input branches, three investor outputs



Input: five sub-scores as logits, log budget and budget band, genre (multi-hot), comp ROI spread, missing-data flags, and the three curves. Missing inputs are set to 0 with their flag set to 1.

Losses: weighted sum of breakeven, ROI range and breakout losses, starting at 1 : 1 : 0.5. Breakout loss is class weighted. The ROI head builds the 50th percentile as the 20th plus a positive step, and the 80th as the 50th plus a positive step, so the range never crosses.

Version 2: genre gating. A genre embedding weights the sub-scores, so the model learns, for example, that market timing matters more for horror.

Why this design. Flat scores suit a small feedforward network; a transformer adds cost without benefit. A CNN fits the curves because their shape carries the signal. The network is small so it can learn from thousands of films without memorizing them.

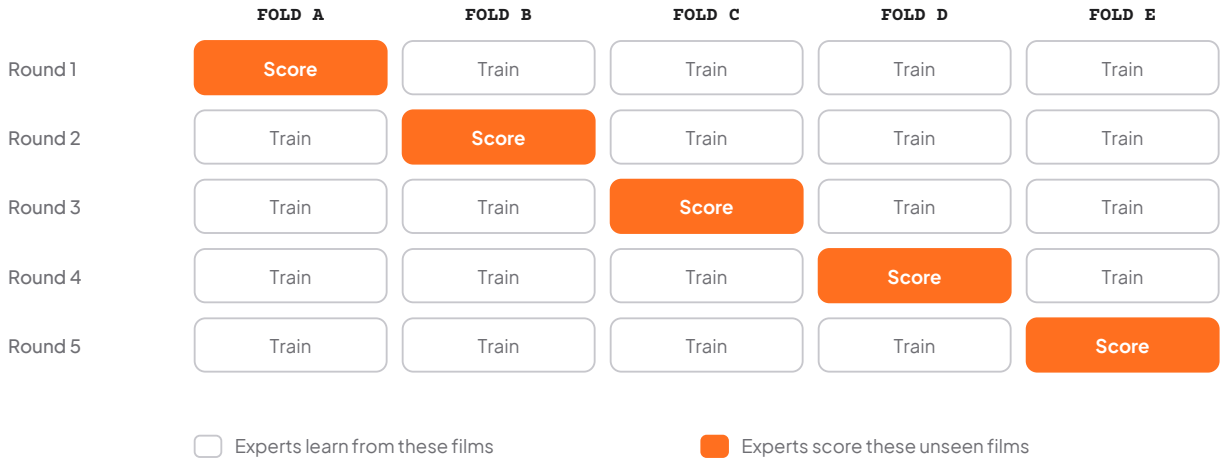
SECTION 14

Training rules

Rule 1: out-of-fold sub-scores. Experts and combiner train on the same films. If an expert scores a film it trained on, the combiner learns to over-trust it.

Out-of-fold scoring

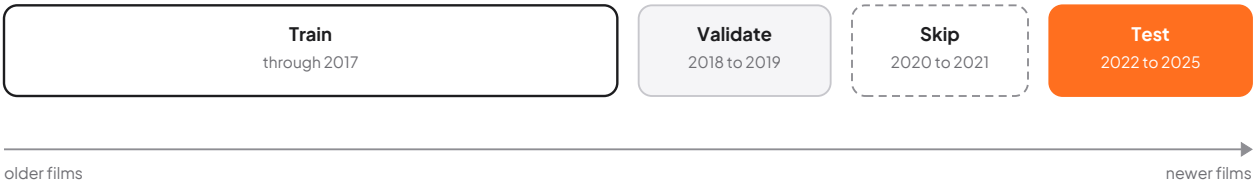
Every sub-score comes from an expert that never saw the film



Rule 2: split by release year, never randomly. In real use the model always predicts films that come after everything it learned from.

Split by release year

The model is always tested on films after everything it learned from



Rule 3: calibrate. A predicted 70% must break even about 70% of the time. Apply temperature scaling on validation; check that actual ROI falls below the 20th percentile prediction about 20% of the time.

Settings: Adam, learning rate 1e-3, weight decay 1e-4, batch 256, early stopping with patience 20, five seeds averaged. Their disagreement feeds the risk level.

SECTION 15

Evaluation

Question	Metric
Does it rank scripts well?	AUC for breakeven, Spearman correlation for ROI
Are probabilities honest?	Brier score, calibration plot
Is the range honest?	Pinball loss, quantile coverage
Would an investor profit?	Breakeven rate of the top 10% scored films vs the overall rate
Is it comparable to competitors?	Box office error on a fixed set of known releases, reported as a percentage

Baselines: logistic regression on genre and budget only, and gradient boosting on the combiner's inputs. If the network does not beat gradient boosting, ship the simpler model.

Ablations: retrain without each part (each sub-score, curves, budget, each LLM block's features). The interview ablation is the evidence for FilmCrowd's proprietary claim.

SECTION 16

Lessons from Largo.ai

Largo.ai, a Swiss EPFL spin-off launched as a SaaS platform in 2020, is the closest comparable product. It reports scene-by-scene emotional analysis, genre fluctuation across the story, character and casting analysis, similar films, budget analysis, and box office and streaming forecasts. Reviewing it found six gaps, now built into this plan.

Gap	What Largo offers	What we added
Talent	Casting recommendations drawing on a large actor-profile database; revenue estimates that use actor fame	Stage E: cast scenarios, talent sub-score
Simulated audiences	Automated focus-group testing with simulated audiences	Stage F: Audience simulator LLM
Audience segments	Audience forecasts and engagement by segment	Segment appeal output
Streaming	Separate box office and streaming forecasts	Streaming label and best-channel output
Fix suggestions	Which scenes to change for a target audience	Report writer LLM with scene-level fixes
Genre per scene	Genre mix measured through the storyline	Genre-mix curve for the CNN

Where FilmCrowd is ahead. The proprietary interview rubric, which public data cannot supply. Calibrated breakeven probability and ROI range rather than a single estimate. The budget curve. Out-of-fold and time-split rigor with a published test metric.

Benchmark to plan for. Largo publicizes cases such as a $201M estimate for Venom against $213M actual, and claims 70% to 80% accuracy. These are company claims, not independent tests, but investors will compare, so our evaluation reports a directly comparable error figure.

SECTION 17

Worked example: Spider-Man (1993 draft)

One real script through the pipeline: the 1993 second revision of Spider-Man, never produced. That is FilmCrowd's situation exactly: a script with no film behind it. The as-of date is 1993, so market and comps use only earlier films.

How to read it. The parsed numbers were computed from the actual script. Everything else is illustrative, since no model is trained yet.

Stage A: parsed from the script (real)

Feature	Value	Signal
Scenes	213	Fast-moving, many short scenes
Interior / exterior	125 / 88 (59% / 41%)	High exterior share: costlier
Night / day	186 / 22 (87% night)	Very night-heavy: slow, expensive shooting
Unique locations	124	Large location load
Speaking characters	46	Mid-sized cast
Dialogue cues	Peter 176, Ock 136, Liz 120, Spider-Man 113	Strong lead, strong villain
Length	About 25,500 words	Roughly two hours

Stage A: LLM outputs (illustrative)

Position	Beat	Emotion
22%	Spider bite during the lab overload	Rising
28%	Peter discovers his powers	Up
37%	Wrestling match debut	Up
52%	Uncle Ben is shot	Sharp fall
64%	Rescue of Liz	Up
82%	Thrown into the well	Lowest point
90% to 98%	Final fight, rescue, romantic ending	Triumphant

Rating (1 to 5)	Score	Budget driver (1 to 5)	Score
Structure	3	VFX	5
Character arc	4	Stunts	5
Villain strength	5	Exteriors and nights	5
Dialogue	3	Locations	4
Tone consistency	3	Sets	4
Entertainment payoff	4	Cast size	3

Budget band: high to tentpole, driven by VFX and night exteriors.

Stages B to F and the investor view (illustrative)

Stage	Result
Market (as of 1993)	Superhero genre proven by Batman (1989) and Batman Returns (1992); rising momentum, low saturation. Sub-score 78.
Interview (placeholder criteria)	Villain is personal: 5. Loss drives the hero: 4, but late. Hero's goal clear early: 3. Sub-score 64.
Comps (approximate)	TMNT (1990) about 15x, Batman (1989) about 9x, Superman (1978) about 5x, Dick Tracy (1990) and Batman Returns (1992) about 3x, The Rocketeer (1991) about 1x. Strong average, very wide spread. Sub-score 74.
Script	Strong villain and payoff, late emotional turn. Sub-score 71.
Talent	Not scored: no cast attached in 1993.
Audience simulator	Strongest with teens and young adults; weaker with older audiences put off by broad comedy.
Investor view	Breakeven 62%. ROI 1.0x / 2.9x / 6.5x. Breakout 28%. Risk high. Best channel: wide theatrical. Budget curve: about 74% at mid budget, 62% at high, 48% at tentpole.
Report writer notes	Risk sits in cost, not story. Control VFX scope and night shoots. Move Uncle Ben's death earlier to tighten the first act.

SECTION 18

Deliverables

Phase 1: data and definitions

- Feature dictionary: name, group, definition, source, pre-release yes/no, supporting interviews
- Success labels, including streaming, computed for the dataset
- Fixed train, validation and test split by release year
- EDA notebook

Phase 2: extraction

- Scene parser, plus Scene tagger, Script rater and Segment scorer LLM prompts with written anchors
- Budget estimator checked against known budgets
- Interview rubric, reviewed by a person
- News reader, market features, comps search
- Human-rated set of 50 scripts to check LLM ratings

Phase 3: expert models

- Four gradient boosting experts with out-of-fold sub-scores

Phase 4: combiner

- Multi-head MLP with CNN curve branch
- Calibration
- Baselines and ablations, including the interview proof

Phase 5: investor view

- Single-script report with all outputs and Report writer notes
- Investor headline metric on the test set

Phase 6: version 2

- Talent scenarios
- Audience simulator, validated against real audience scores
- Streaming forecast and best-channel output
- Genre gating

SECTION 19

Risks and open questions

Risk	Mitigation
Few films have script, budget and outcome together	Synopses for coverage, full scripts as a high-quality subset; missing flags
Noisy or missing budgets	Train money heads only on known budgets; report coverage
LLM ratings drift between runs or prompt versions	Written anchors, three runs averaged, versioned prompts, human-rated check set
Simulated audiences sound plausible but are wrong	Use as a feature only after validation against real audience scores
Data licensing	Fine for research; review before the product ships
Correlation, not cause	Present findings as associations

Open questions

- Theatrical, streaming or independent: which path should the labels target first?
- How many films in the 1M dataset have full scripts and known budgets?
- What format are the interview transcripts in, and how many are there?

Sources on Largo.ai

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- EPFL news, 75th Cannes Film Festival pitch session: actu.epfl.ch/news/it-s-an-amazing-feeling-to-be-at-the-75th-cannes-4
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